Stage 8 Maths — Practice Test 2 — Mark Scheme

Paper 1: 60 minutes · no calculator · Paper 2: 60 minutes · calculator allowed · Total 100 marks

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 60 minutes — 50 marks

1 **[1]**

(a) Work out $20 - 4 \times 3^2 \div 6$

Answer: 14
Notes / working: Indices first: $3^2 = 9$. Then × and ÷ from left to right: $4 \times 9 = 36$, $36 \div 6 = 6$. Subtract last: $20 - 6 =$ **14**. (Working left to right gives $16 \times 9 \div 6 = 24$, which is wrong.)

(b) Brackets are added to $5 + 3 \times 4 - 2$ so that the answer is 30. Which one is correct? **[1]**

- ☐ A) $(5 + 3) \times 4 - 2$
- ☐ B) $5 + (3 \times 4) - 2$
- ☐ C) $5 + 3 \times (4 - 2)$
- ☐ D) $(5 + 3) \times (4 - 2)$

Answer: A) $(5 + 3) \times 4 - 2$
Notes / working: $(5 + 3) \times 4 - 2 = 8 \times 4 - 2 = 32 - 2 =$ **30** ✓. The others give 15, 11 and 16.

2 **[1]**

(a) Work out $(-84) \div (-4) - 3 \times (-5)$

Answer: 36
Notes / working: $(-84) \div (-4) = 21$ (negative ÷ negative = positive). $3 \times (-5) = -15$. So $21 - (-15) = 21 + 15 =$ **36**.

(b) Ali says: “When you multiply three negative integers together, the answer is always negative.” **[1]**
Is Ali correct? Explain how you know.

Answer: 1 mark: Yes (correct) with a correct reason, e.g. negative × negative = positive, then positive × negative = negative; or a general argument using signs. Accept a correct example with the sign rule stated, e.g. $(-2) \times (-3) \times (-4) = 6 \times (-4) = -24$. Do not accept a single example with no reason (one example does not show it is ALWAYS true). Do not accept ‘No’.
Notes / working: Yes. Multiply the first two: negative × negative = positive. Then positive × the third negative = negative. This works for any three negative integers, e.g. $(-2) \times (-3) \times (-4) = 6 \times (-4) = -24$.

3 The Hassan family drive from Cairo to Alexandria for their summer holiday. The road distance is about 200 km. **[1]**

Use 1 km ≈ 5/8 of a mile to find the distance in miles.

> **Answer:** 125 miles
> **Notes / working:** 200 × 5/8 = 200 ÷ 8 × 5 = 25 × 5 = **125 miles**. Check: 1 mile ≈ 1.6 km, and 125 × 1.6 = 200 ✓.

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4 On holiday the Hassan family drink 2 3/4 litres of bottled water each day. **[2]**

(a) How many litres do they drink in 6 days? Give your answer as a mixed number.

> **Answer:** 16.5 or 33/2 litres
> **Notes / working:** Estimate first: 6 × 3 = 18, so the answer is a bit less than 18. 6 × 2 3/4 = 6 × 2 + 6 × 3/4 = 12 + 18/4 = 12 + 4 1/2 = **16 1/2 litres**.
> Mark scheme: 2 marks for 16 1/2 (oe, e.g. 33/2 or 16.5). 1 mark for 12 + 18/4 or 6 × 11/4 = 66/4 seen.

(b) They pour a 9-litre bottle of water into cups. Each cup holds 3/4 of a litre. How many cups can they fill? **[1]**

> **Answer:** 12
> **Notes / working:** 9 ÷ 3/4 = 9 × 4/3 = 36/3 = **12 cups**. (There are 4 quarters in each litre, so 36 quarters, and 36 ÷ 3 = 12.)

(c) On Monday the family walk 4 1/3 km along the Corniche. On Tuesday they walk 1 3/4 km. How much further do they walk on Monday? Give your answer as a mixed number in its simplest form. **[2]**

> **Answer:** 2 7/12 or 2 7/12 km
> **Notes / working:** Estimate: 4 − 2 = 2, so about 2 or 3 km. 4 1/3 − 1 3/4 = 13/3 − 7/4 = 52/12 − 21/12 = 31/12 = **2 7/12 km**. (Or: 4 1/3 − 1 = 3 1/3, then 3 4/12 − 9/12 = 2 16/12 − 9/12 = 2 7/12.)
> Mark scheme: 2 marks for 2 7/12 cao. 1 mark for a correct common denominator seen, e.g. 52/12 − 21/12 or 4/12 and 9/12, or for 31/12 not written as a mixed number.

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5 A hotel room in Alexandria cost E£800 a night last summer. This summer the same room costs E£920 a night. **[1]**

(a) Work out the absolute change in the price.

> **Answer:** 120 E£
> **Notes / working:** Absolute change = new − old = 920 − 800 = **E£120** (an increase).

(b) Work out the percentage increase in the price. **[2]**

> **Answer:** 15 %
> **Notes / working:** Percentage increase = change ÷ original × 100 = 120 ÷ 800 × 100. 120/800 = 12/80 = 3/20 = 15/100, so **15%**.
> Mark scheme: 2 marks for 15. 1 mark for 120/800 (oe) seen. Do not divide by the new price: 120 ÷ 920 is wrong.

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6 **[1]**

(a) $5^7 \times 5^2 \div 5^4 = 5^n$. Find the value of n.

> **Answer:** 5
> **Notes / working:** Multiply → add the indices; divide → subtract: 7 + 2 − 4 = 5, so **n = 5**.

(b) Write down the value of $\sqrt[3]{(-64)}$. **[1]**

> **Answer:** -4
> **Notes / working:** (−4) × (−4) × (−4) = 16 × (−4) = −64, so $\sqrt[3]{(-64)}$ = **−4**. A negative number has a negative cube root.

(c) Work out $(-5)^2 - (-2)^3$ **[1]**

> **Answer:** 33
> **Notes / working:** $(-5)^2$ = 25 (negative squared is positive). $(-2)^3$ = −8 (negative cubed is negative). 25 − (−8) = 25 + 8 = **33**.

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7 **[1]**

Work out 0.72 ÷ 0.01

> **Answer:** 72
> **Notes / working:** Dividing by 0.01 is the same as multiplying by 100: 0.72 × 100 = **72**.

8 **[1]**

(a) Which fraction is equal to a recurring decimal?

- ☐ A) 3/8
- ☐ B) 7/20
- ☐ C) 5/12
- ☐ D) 9/25

Answer: C) 5/12
Notes / working: 3/8 = 0.375, 7/20 = 0.35 and 9/25 = 0.36 all stop (terminate). **5/12** = 0.41666…, the 6 recurs. A fraction in its simplest form terminates only when the denominator has no prime factors other than 2 and 5; 12 = 2 × 2 × 3 has a factor 3.

(b) Zara says: “7/30 is a terminating decimal, because 30 is a multiple of 10.” **[1]**
Is Zara correct? Explain your answer.

Answer: 1 mark: No, with a correct reason, e.g. 7/30 = 0.2333… (the 3 recurs); or 30 = 2 × 3 × 5 has a prime factor 3 (7/30 is already in simplest form); or 7/30 = 0.7 ÷ 3 = 0.2333… Do not accept ‘No’ with no reason or a wrong reason.
Notes / working: Zara is **not correct**. 7/30 = 0.7 ÷ 3 = 0.2333…, so it recurs. 30 = 2 × 3 × 5, and the factor 3 makes the decimal recur. Being a multiple of 10 is not enough.

(c) Which list is in order, starting with the smallest? **[1]**

- ☐ A) −0.6, −5/8, −0.65, −2/3
- ☐ B) −0.65, −2/3, −0.6, −5/8
- ☐ C) −5/8, −0.6, −2/3, −0.65
- ☐ D) −2/3, −0.65, −5/8, −0.6

Answer: D) −2/3, −0.65, −5/8, −0.6
Notes / working: Write as decimals: −2/3 = −0.666…, −0.65, −5/8 = −0.625, −0.6. For negative numbers the one furthest from zero is the smallest: **−2/3 < −0.65 < −5/8 < −0.6**. The first list is the right order but largest first.

9 **[2]**

Work out 8.64 ÷ 0.4

Answer: 21.6
Notes / working: Multiply both numbers by 10 so you divide by a whole number: 86.4 ÷ 4 = **21.6**.
Estimate: 8 ÷ 0.4 = 20 ✓.
Mark scheme: 2 marks for 21.6. 1 mark for 86.4 ÷ 4 seen, or for an answer with the correct digits 216 (e.g. 2.16 or 216).

10 **[1]**

Show how to work out 3.7 × 46 + 3.7 × 54 without a calculator. Write down the answer.

Answer: 1 mark: 370 with a method using the distributive law, e.g. 3.7 × (46 + 54) = 3.7 × 100 = 370. Accept 370 from correct long multiplications (170.2 + 199.8 = 370).
Notes / working: 3.7 is a common factor: 3.7 × 46 + 3.7 × 54 = 3.7 × (46 + 54) = 3.7 × 100 = **370**.

11 **[1]**

(a) Which one of these is a formula?

- ☐ A) 4x + 3
- ☐ B) 4x + 3 = 19
- ☐ C) x > 4
- ☐ D) P = 4s, where P is the perimeter of a square of side s

Answer: D) P = 4s, where P is the perimeter of a square of side s
Notes / working: P = 4s is a formula: it connects two quantities and works for every value of s. 4x + 3 is an expression (no = sign). 4x + 3 = 19 is an equation (true for one value, x = 4). x > 4 is an inequality.

(b) Simplify $3a^2 + 5a - a^2 + 2a$ **[1]**

Answer: 2a^2+7a or 7a+2a^2 or 2a²+7a or 7a+2a²
Notes / working: Collect like terms: $3a^2 - a^2 = 2a^2$; 5a + 2a = 7a. Answer **$2a^2 + 7a$**. a^2 and a are not like terms, so they cannot be added. Type a^2 as a^2.

(c) Expand $2x(x^2 - 3)$ **[1]**

Answer: 2x^3-6x or -6x+2x^3 or 2x³-6x or -6x+2x³
Notes / working: $2x \times x^2 = 2x^3$; $2x \times (-3) = -6x$. Answer **$2x^3 - 6x$**. Type it as 2x^3 − 6x.

(d) Factorise completely $12y^2 + 18y$ **[2]**

Answer: 6y(2y+3) or 6y(3+2y) or (2y+3)6y or (3+2y)6y or 6y×(2y+3)
Notes / working: The HCF of $12y^2$ and 18y is 6y. $12y^2 \div 6y = 2y$ and $18y \div 6y = 3$, so **6y(2y + 3)**. Check by expanding: $12y^2 + 18y$ ✓.
Mark scheme: 2 marks for 6y(2y + 3). 1 mark for a correct partial factorisation, e.g. $6(2y^2 + 3y)$, 3y(4y + 6) or 2y(6y + 9).

12 **[2]**

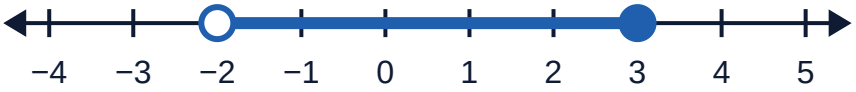
(a) Solve 5x − 7 = 2x + 11

Answer: Answer x = 6. 2 marks for x = 6 with working. 1 mark for a correct first step, e.g. 3x − 7 = 11, 5x = 2x + 18 or 3x = 18. Accept x = 6 from trial and improvement only if a check is shown (2 marks). Do not award marks for 6 with wrong working (nfww).
Notes / working: Subtract 2x from both sides: 3x − 7 = 11. Add 7: 3x = 18. Divide by 3: **x = 6**. Check: 5 × 6 − 7 = 23 and 2 × 6 + 11 = 23 ✓.

(b) Solve (x + 4)/3 = 5 **[1]**

Answer: 11
Notes / working: Multiply both sides by 3: x + 4 = 15. Subtract 4: **x = 11**.

[1]

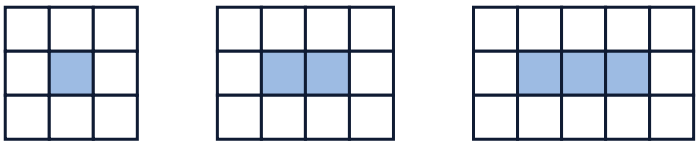


(c) The diagram shows an inequality on a number line.
Which inequality is shown?

- ☐ A) −2 ≤ x < 3
- ☐ B) −2 < x ≤ 3
- ☐ C) −2 ≤ x ≤ 3
- ☐ D) −2 < x < 3

Answer: B) −2 < x ≤ 3
Notes / working: The open (empty) circle at −2 means −2 is **not** included, so −2 < x. The filled circle at 3 means 3 **is** included, so x ≤ 3. Answer: **−2 < x ≤ 3**.

13 Rashid makes patterns with square tiles. Each pattern has a row of blue tiles with white tiles all round it. **[1]**



(a) How many white tiles are in Pattern 5?

Answer: 16
Notes / working: White tiles: 8, 10, 12, … The term-to-term rule is +2. Pattern 4 has 14, Pattern 5 has **16**. (Or: Pattern 5 is 3 rows of 7 tiles = 21, minus 5 blue = 16.)

(b) Write an expression for the number of white tiles in Pattern n. **[2]**

> **Answer:** 2n+6 or 6+2n or 2(n+3) or 2×n+6 or 2n+6 tiles
> **Notes / working:** The sequence goes up by 2 each time, so it starts 2n. When n = 1, 2n = 2, but there are 8 tiles, so add 6: **2n + 6**. Check n = 3: 2 × 3 + 6 = 12 ✓. (Picture reason: 3 rows of n + 2 tiles is 3n + 6 tiles, minus the n blue tiles = 2n + 6.)
> Mark scheme: 2 marks for 2n + 6 (oe). 1 mark for 2n + c where c ≠ 6, or for 2n seen.

(c) Rashid says one of his patterns has exactly 75 white tiles. Explain why he cannot be correct. **[1]**

> **Answer:** 1 mark for a correct reason, e.g. 2n + 6 is always even and 75 is odd; or 2n + 6 = 75 gives n = 34.5, which is not a whole number; or the pattern numbers next to 75 have 74 (n = 34) and 76 (n = 35) white tiles. Do not accept '75 is too big'.
> **Notes / working:** The number of white tiles 2n + 6 = 2(n + 3) is always **even**, but 75 is odd. (Or: 2n + 6 = 75 gives 2n = 69, n = 34.5, which is not a pattern number.)

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14 **[1]**

(a) Here is a function machine:
input → × 3 → − 4 → output
The output is 17. Work out the input.

> **Answer:** 7
> **Notes / working:** Work backwards with inverse operations: 17 + 4 = 21, then 21 ÷ 3 = **7**. Check: 7 × 3 − 4 = 17 ✓.

(b) Which line is parallel to y = 3x − 2? **[1]**

- ☐ A) y = 2x − 3
- ☐ B) y = 3x + 5
- ☐ C) y = −3x − 2
- ☐ D) y = x + 3

> **Answer:** B) y = 3x + 5
> **Notes / working:** Parallel lines have the same gradient m. y = 3x − 2 has gradient 3, and so does **y = 3x + 5**. y = −3x − 2 has the same y-intercept (−2), not the same gradient.

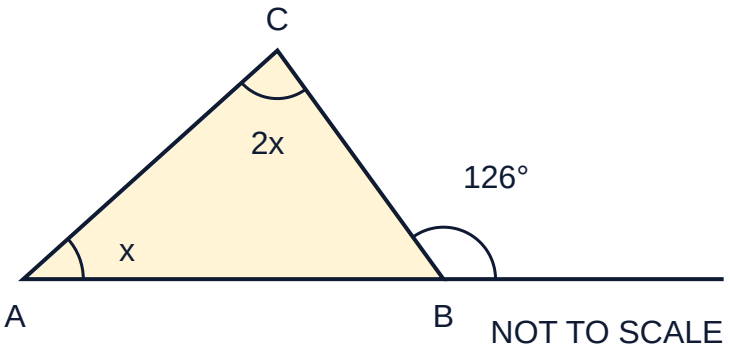
15 Omar is drawing a tiling pattern. He says: “A rhombus is a special kind of square.” **[1]**

Explain why Omar is wrong.

Answer: 1 mark for a correct explanation, e.g. a rhombus does not need right angles (a square must have four right angles); OR it is the other way round: every square is a rhombus, but not every rhombus is a square. Accept a sketch of a rhombus without right angles with a correct comment.
Notes / working: A square has 4 equal sides **and** 4 right angles. A rhombus only needs 4 equal sides; its angles do not have to be 90°. So every square is a special rhombus, but a rhombus is not always a square. Omar has it the wrong way round.

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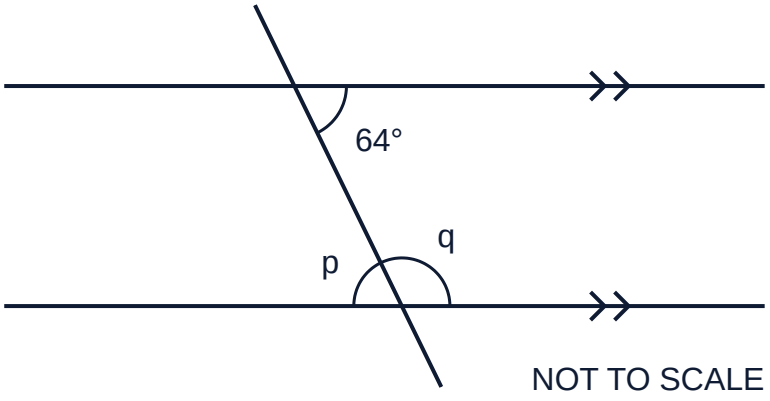
16 In triangle ABC, AB is extended. Angle A = x and angle C = 2x. The exterior angle at B is 126°. **[2]**



Work out the value of x.

Answer: Answer x = 42. 2 marks for 42. 1 mark for x + 2x = 126 (or 3x = 126) seen, OR for angle ABC = 54° with x + 2x + 54 = 180 (oe). Do not accept 42 from measuring the diagram (it is not to scale).
Notes / working: The exterior angle of a triangle equals the sum of the two interior opposite angles: x + 2x = 126, so 3x = 126 and **x = 42°**. Check: angles 42°, 84° and 180° − 126° = 54° add to 180° ✓.

17 The diagram shows two parallel lines crossed by a straight line. **[1]**



(a) Work out the size of angle q.

Answer: 116 °
Notes / working: p = 64° (alternate angles). p and q lie on a straight line, so q = 180° − 64° = **116°**.

(b) Angle p = 64°. Give a geometrical reason. **[1]**

Answer: 1 mark for ‘alternate angles (are equal)’. Accept a correct two-step reason, e.g. corresponding angles with vertically opposite angles. Do not accept ‘Z angles’ alone, or ‘they look equal’.
Notes / working: p and the 64° angle are on opposite sides of the crossing line, between the parallel lines (a Z shape): they are **alternate angles**, so they are equal.

18 Use a ruler and compasses only. Do not rub out your construction arcs. **[2]**

Construct triangle ABC with AB = 7 cm, AC = 5 cm and BC = 6 cm.

Answer: 2 marks: an accurate triangle (all three sides correct to within 2 mm) with both pairs of construction arcs seen. 1 mark: an accurate triangle without arcs, OR correct arcs (radius 5 cm from A and 6 cm from B, or the other way round) with one side wrong by more than 2 mm.
Notes / working: 1) Draw AB = 7 cm with a ruler. 2) Open the compasses to 5 cm, put the point on A and draw an arc above AB. 3) Open the compasses to 6 cm, put the point on B and draw an arc that crosses the first arc. 4) Where the arcs cross is C. Join AC and BC. Check with a ruler: AC = 5 cm, BC = 6 cm. Leave the arcs — they show your method.

19 **[1]**

(a) A boat sails from the harbour in Alexandria on a bearing of 065° to a fishing area.
Work out the bearing of the harbour from the fishing area.

Answer: 245 °
Notes / working: The back bearing is 180° more (because 065° is less than 180°): 65 + 180 = **245°**.

(b) On a plan of a garden, a straight path goes from P(−3, 4) to Q(5, −2). A bench is at the midpoint of PQ. **[1]**
Work out the coordinates of the bench.

Answer: (1,1) or 1,1
Notes / working: Midpoint = (mean of the x-coordinates, mean of the y-coordinates) = ((−3 + 5)/2, (4 + (−2))/2) = (2/2, 2/2) = **(1, 1)**.

20 The school canteen sells a meal deal. Students choose one main and one drink. **[1]**
Mains: rice, pasta, sandwich
Drinks: water, juice, milk, lemonade

(a) List all the possible meal deals, or show them in a table.

Answer: 1 mark for all 12 combinations listed or shown in a correct table (3 × 4), with no repeats. Abbreviations (e.g. R/W, P/J) are fine.
Notes / working: Use a table: rows rice, pasta, sandwich; columns water, juice, milk, lemonade. 3 × 4 = **12** meal deals: RW, RJ, RM, RL, PW, PJ, PM, PL, SW, SJ, SM, SL.

(b) Laila chooses a main and a drink at random. Find the probability that she chooses pasta and juice. **[1]**

Answer: 1/12
Notes / working: There are 12 equally likely meal deals and only 1 is pasta with juice, so P = **1/12**.

(c) The probability that the canteen sells out of sandwiches on a school day is 0.15. **[1]**
Find the probability that the canteen does not sell out of sandwiches on a school day.

Answer: 0.85 or 17/20
Notes / working: ‘Sells out’ and ‘does not sell out’ are complementary events, so their probabilities add to 1: 1 − 0.15 = **0.85**.

Paper 2 — 60 minutes — 50 marks

1 Uncle Sami visits from London for the Hassan family's holiday in Alexandria. On the day he changes money, £1 = E£61.40. **[1]**

(a) Uncle Sami changes £250 into Egyptian pounds. How many Egyptian pounds does he get?

Answer: 15350 E£
Notes / working: 250 × 61.40 = **E£15 350**.

(b) A boat trip costs E£450 per person. Uncle Sami pays for 5 people. Work out the total cost in pounds (£). Give your answer to the nearest penny. **[2]**

Answer: 36.64 £
Notes / working: Total in Egyptian pounds: 5 × 450 = E£2250. Change to pounds: 2250 ÷ 61.40 = 36.6449… = **£36.64** to the nearest penny.
Mark scheme: 2 marks for 36.64. 1 mark for 2250 ÷ 61.40 (oe) seen, or 36.6(4…) not rounded correctly. Do not multiply by 61.40.

(c) A beach hotel welcomed 38 549 guests last year. **[1]**
Round 38 549 to 3 significant figures.

Answer: 38500
Notes / working: The first three significant figures are 3, 8, 5. The next digit is 4 (less than 5), so round down: **38 500**. Keep the zeros as place holders (385 is wrong).

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2 The price of a train ticket from Cairo to Alexandria goes up from E£85 to E£102. **[2]**

Work out the percentage increase.

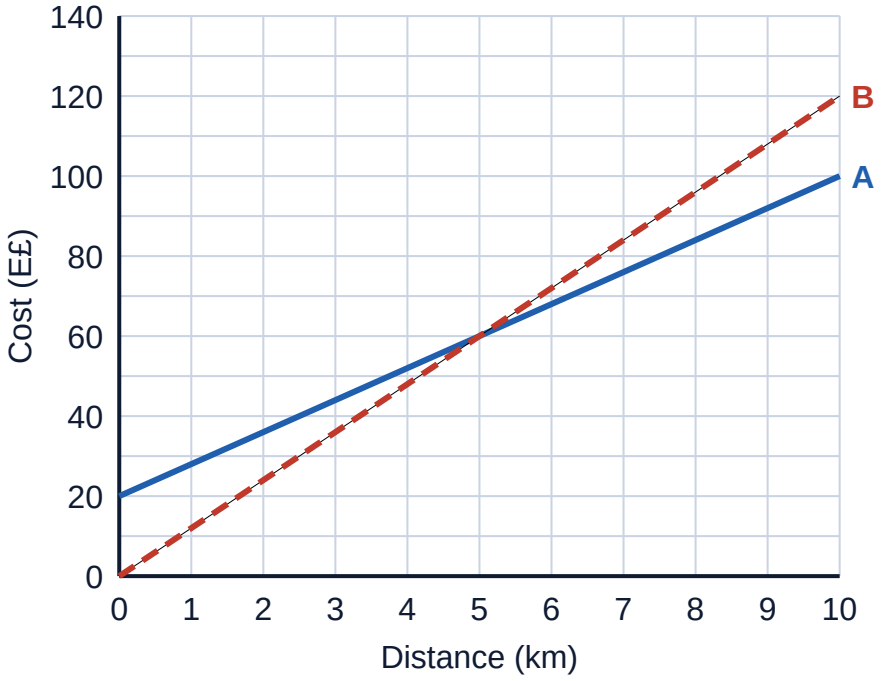
Answer: 20 %
Notes / working: Increase = 102 − 85 = 17. Percentage increase = 17 ÷ 85 × 100 = **20%**. (Or multiplier: 102 ÷ 85 = 1.2, which is a 20% increase.)
Mark scheme: 2 marks for 20. 1 mark for 17 ÷ 85 or 102 ÷ 85 = 1.2 seen.

3 Two taxi companies in Alexandria charge like this: **[1]**

Company A: E£20 plus E£8 for each kilometre

Company B: E£12 for each kilometre

The graph shows the cost of a journey for each company.



(a) Use the graph. How far can you travel with Company A for E£100?

Answer: 10 km

Notes / working: Go across from E£100 to line A, then down: **10 km**. Check: 20 + 8 × 10 = 100 ✓.

(b) For what distance do both companies charge the same amount? **[1]**

Answer: 5 km

Notes / working: The lines cross at (5, 60): for a **5 km** journey both charge E£60. Check: 20 + 8 × 5 = 60 and 12 × 5 = 60 ✓. For shorter journeys B is cheaper; for longer journeys A is cheaper.

(c) Mona takes a 9 km journey. Which company is cheaper, and by how much? Show your working. **[2]**

Answer: 1 mark for both costs: A = 20 + 8 × 9 = E£92 and B = 12 × 9 = E£108 (accept values read from the graph within E£2). 1 mark for 'Company A, by E£16' (FT their two costs if the company and difference are correct for them).

Notes / working: Company A: 20 + 8 × 9 = 20 + 72 = E£92. Company B: 12 × 9 = E£108. **Company A is cheaper, by 108 − 92 = E£16.**

(d) For Company A, the cost C (E£) for d km is C = 20 + 8d. **[1]**
Make d the subject of the formula C = 20 + 8d.

> **Answer:** 1 mark for d = (C − 20)/8 oe, e.g. d = (C − 20) ÷ 8 or d = C/8 − 2.5. Do not accept d = C − 20/8 without brackets.
> **Notes / working:** C = 20 + 8d. Subtract 20: C − 20 = 8d. Divide by 8: **d = (C − 20)/8**.

(e) The Hassan family also compare phone plans. Plan Z costs a fixed E£30 each month plus E£5 for each GB of data used. **[1]**
y is the monthly cost in E£ and x is the number of GB used. Write an equation for y in terms of x, in the form y = mx + c.

> **Answer:** y=5x+30 or y=30+5x or y=5×x+30
> **Notes / working:** The amount per GB is the gradient (m = 5) and the fixed charge is the y-intercept (c = 30): **y = 5x + 30**.

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4 A school cycling club takes part in a race. Karim's bicycle wheel has a diameter of 68 cm. **[2]**

(a) Work out the circumference of the wheel. Give your answer correct to 1 decimal place.

> **Answer:** 213.6 cm
> **Notes / working:** C = πd = π × 68 = 213.628… = **213.6 cm** (1 d.p.).
> Mark scheme: 2 marks for 213.6 (accept 213.5 to 213.7 from a value of π like 3.142). 1 mark for π × 68 seen, or for 213.(6…) not rounded correctly. Do not use the radius formula with d (2 × π × 68 = 427.3 is wrong).

(b) One stage of the race is 5 km long. How many complete turns does the wheel make in this stage? **[2]**

> **Answer:** Answer 2340. 2 marks for 2340. 1 mark for 500 000 ÷ their circumference seen (e.g. 500 000 ÷ 213.6), OR for 2340.5… or 2341 as the answer. The units must match: using 5000 (metres) with the circumference in cm scores 0.
> **Notes / working:** 5 km = 5 × 1000 × 100 = 500 000 cm. 500 000 ÷ 213.628… = 2340.5… So the wheel makes **2340 complete turns** (round down: the 2341st turn is not finished).

(c) On a track, Rider P completes a lap every 6 minutes and Rider Q completes a lap every 8 minutes. They start together on the start line. **[1]**
After how many minutes are they next on the start line together?

> **Answer:** 24 minutes
> **Notes / working:** Multiples of 6: 6, 12, 18, **24**, … Multiples of 8: 8, 16, **24**, … The lowest common multiple is **24 minutes**.

5 The race times, in minutes, of the 15 riders in the Red team are shown in the stem-and-leaf diagram. **[1]**

Stem	Leaf
3	8
4	1 3 7
5	0 2 2 5 8
6	0 1 4 9
7	2 5

Key: a stem of 4 with a leaf of 1 means 41 minutes

(a) Find the median time of the Red team.

Answer: 55 minutes
Notes / working: There are 15 times, so the median is the (15 + 1) ÷ 2 = 8th value. In order: 38, 41, 43, 47, 50, 52, 52, **55**, … Median = **55 minutes**.

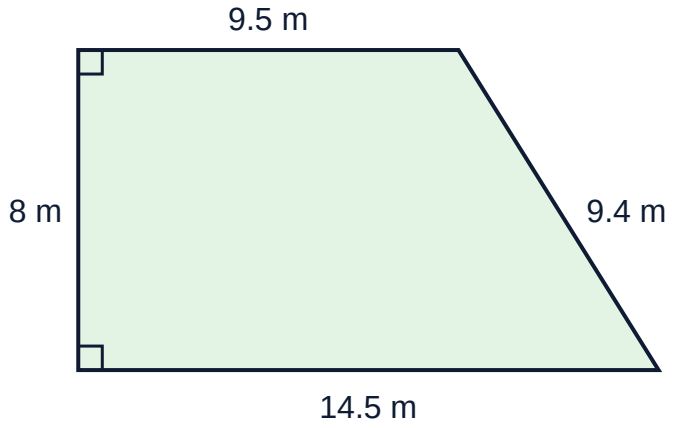
(b) Find the range of the Red team's times. **[1]**

Answer: 37 minutes
Notes / working: Range = largest − smallest = 75 − 38 = **37 minutes**.

(c) The Blue team's times have a median of 58 minutes and a range of 21 minutes. **[2]**
Compare the times of the two teams. Write two comparisons.

Answer: 1 mark for comparing averages in context: the Red team was faster on average / had lower times because its median (55) is lower than Blue's (58). 1 mark for comparing spread in context: the Blue team's times were more consistent (less spread out) because its range (21) is smaller than Red's (37). Comparisons must use the statistics, not single riders. Do not accept ‘Red is better’ without a reason.
Notes / working: 1) **Average:** Red's median (55 min) is lower than Blue's (58 min), so the Red riders were faster on average (a lower time is better in a race). 2) **Spread:** Blue's range (21 min) is smaller than Red's (37 min), so the Blue riders' times were more consistent.

6 Nour is designing a garden. The garden is a trapezium. **[2]**



(a) Work out the area of the garden.

Answer: 96 m^2
Notes / working: A = ½(a + b)h = ½ × (14.5 + 9.5) × 8 = ½ × 24 × 8 = **96 m²**. Use the perpendicular height (8 m), not the sloping side (9.4 m).
Mark scheme: 2 marks for 96. 1 mark for ½ × (14.5 + 9.5) × 8 seen.

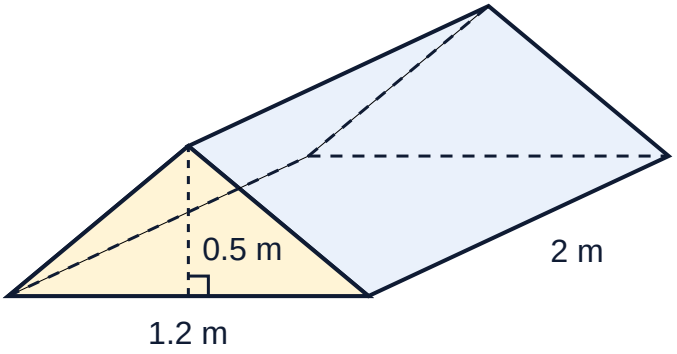
(b) Nour puts fencing all the way round the garden, except for a gate 3 m wide. Fencing is sold in panels 1.8 m long. **[2]**
How many panels does Nour need to buy?

Answer: Answer 22. 2 marks for 22. 1 mark for 38.4 (fencing needed) seen, OR for 21.3(3…) seen. Do not accept 21 (not enough fencing). 23 (from 41.4 ÷ 1.8, gate not taken off) scores 0.
Notes / working: Perimeter = 14.5 + 9.4 + 9.5 + 8 = 41.4 m. Fencing needed = 41.4 − 3 = 38.4 m. 38.4 ÷ 1.8 = 21.33… Round **up** (21 panels are not enough): **22 panels**.

(c) Inside the garden, a flowerbed is a parallelogram with base 3.6 m and perpendicular height 2.5 m. **[1]**
Work out the area of the flowerbed.

Answer: 9 m^2
Notes / working: A = base × perpendicular height = 3.6 × 2.5 = **9 m²**.

[2]



(d) A rockery is a triangular prism made of soil. The triangle has base 1.2 m and height 0.5 m. The prism is 2 m long.
Work out the volume of soil in the rockery.

Answer: 0.6 m^3
Notes / working: Area of the triangle = ½ × 1.2 × 0.5 = 0.3 m². Volume = area of cross-section × length = 0.3 × 2 = **0.6 m³**.
Mark scheme: 2 marks for 0.6. 1 mark for ½ × 1.2 × 0.5 (= 0.3) seen, or 1.2 × 0.5 × 2 = 1.2 (forgot ½).

(e) Nour makes a planter box with no lid. It is a cuboid 80 cm long, 40 cm wide and 30 cm high, made of wood on the base and the four sides. **[2]**
Work out the area of wood needed.

Answer: 10400 cm^2
Notes / working: Base: 80 × 40 = 3200 cm². Front and back: 2 × 80 × 30 = 4800 cm². Two ends: 2 × 40 × 30 = 2400 cm². No lid. Total = 3200 + 4800 + 2400 = **10 400 cm²**.
Mark scheme: 2 marks for 10 400. 1 mark for 13 600 (the lid included) or for three of the five faces correct.

(f) A rectangular vegetable patch is (2x + 3) m long and (x − 1) m wide. Its perimeter is 25 m. **[2]**
Write an equation and solve it to find x.

Answer: Answer x = 3.5. 1 mark for a correct equation, e.g. 2(2x + 3) + 2(x − 1) = 25 or 6x + 4 = 25. 1 mark for x = 3.5 (oe, e.g. 7/2) from correct working.
Notes / working: Perimeter = 2(2x + 3 + x − 1) = 2(3x + 2) = 6x + 4. So 6x + 4 = 25, 6x = 21, **x = 3.5**. Check: the patch is 10 m by 2.5 m and 2 × (10 + 2.5) = 25 ✓.

(g) A garden lamp is a 3D shape with 12 vertices and 18 edges. **[1]**
Use Euler's formula to work out the number of faces.

Answer: 8 faces
Notes / working: Euler's formula: V − E + F = 2 (or F + V − E = 2). 12 − 18 + F = 2, so F = **8**. (This could be a hexagonal prism: 2 hexagons + 6 rectangles.)

7 A tile designer works with square tiles. A rectangular wall panel is 84 cm by 120 cm. It is covered exactly with identical square tiles, with no cutting. The tiles are as large as possible. **[1]**

(a) What is the side length of each tile?

> **Answer:** 12 cm
> **Notes / working:** The side must divide both 84 and 120, and be as large as possible: the HCF. 84 = $2^2 \times 3 \times 7$ and 120 = $2^3 \times 3 \times 5$, so HCF = $2^2 \times 3$ = **12 cm**.

(b) How many tiles cover the panel? **[1]**

> **Answer:** 70 tiles
> **Notes / working:** 84 ÷ 12 = 7 tiles one way and 120 ÷ 12 = 10 tiles the other way: 7 × 10 = **70 tiles**. (Or 84 × 120 ÷ 144 = 10 080 ÷ 144 = 70.)

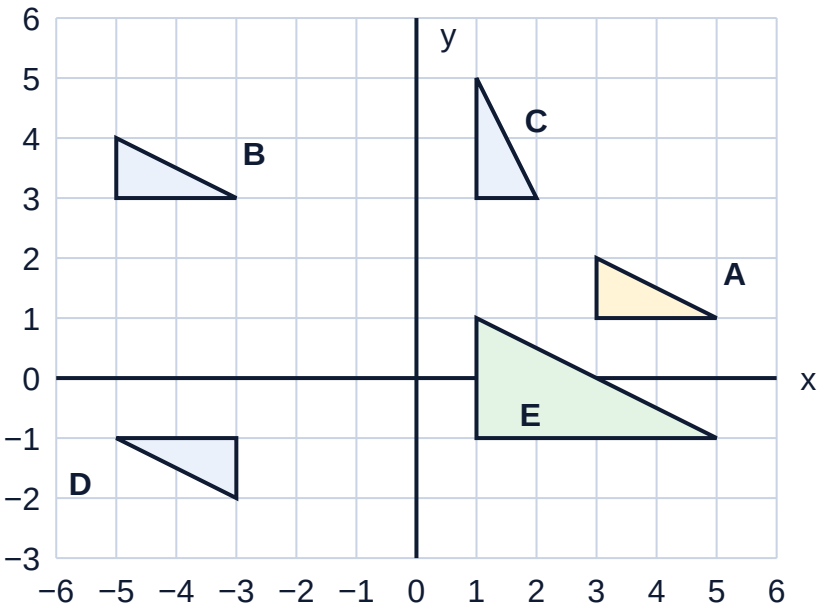
(c) One design uses regular octagons. **[1]**
Write down the order of rotational symmetry of a regular octagon.

> **Answer:** 8
> **Notes / working:** A regular polygon with n sides has n lines of symmetry and rotational symmetry of order n. An octagon has 8 sides, so the order is **8**.

(d) Tiles come in boxes of 24. Hana buys n boxes and uses all the tiles except 7. **[1]**
Write an expression, in terms of n, for the number of tiles Hana uses.

> **Answer:** 24n-7 or -7+24n or 24×n-7 or 24n-7 tiles
> **Notes / working:** n boxes of 24 tiles is 24n tiles. She does not use 7 of them: **24n − 7**.

8 A tile designer draws triangle A and four images of it, B, C, D and E, on a coordinate grid. **[1]**



(a) Triangle A is translated to triangle B. Write down the vector of this translation, as (x, y).

Answer: (-8,2) or -8,2
Notes / working: Pick a corner: the right angle of A is at (3, 1) and of B is at (−5, 3). Move 8 left (−8) and 2 up (+2). Vector **(−8, 2)**, written as a column vector with −8 on top and 2 underneath.

(b) Describe the single transformation that maps triangle A onto triangle C. **[1]**

- ☐ A) Rotation 90° anticlockwise about (0, 0)
- ☐ B) Reflection in the y-axis
- ☐ C) Reflection in the line y = x
- ☐ D) Translation by vector (−2, 2)

Answer: C) Reflection in the line y = x
Notes / working: Each point (a, b) of A goes to (b, a): (3, 1) → (1, 3), (5, 1) → (1, 5), (3, 2) → (2, 3). Swapping the coordinates is a **reflection in the line y = x**. (Rotating 90° anticlockwise about (0, 0) would send (3, 1) to (−1, 3).)

(c) Describe the single transformation that maps triangle A onto triangle D. **[1]**

- ☐ A) Rotation 180° about (0, 0)
- ☐ B) Reflection in the x-axis
- ☐ C) Reflection in the y-axis
- ☐ D) Translation by vector (−8, −2)

Answer: A) Rotation 180° about (0, 0)
Notes / working: (3, 1) → (−3, −1), (5, 1) → (−5, −1), (3, 2) → (−3, −2): both coordinates change sign, so it is a **rotation of 180° about (0, 0)** (the direction does not matter for 180°). A reflection in the x-axis would keep the x-coordinates positive.

(d) Triangle E is an enlargement of triangle A. Write down the scale factor of the enlargement. **[1]**

> **Answer:** 2
> **Notes / working:** The short side of A is 1 unit (from (3, 1) to (3, 2)); the matching side of E is 2 units (from (1, −1) to (1, 1)). The long side goes from 2 to 4 units. Scale factor = **2**.

(e) Find the coordinates of the centre of the enlargement that maps A onto E. **[1]**

> **Answer:** (5,3) or 5,3
> **Notes / working:** Draw straight lines through matching corners: (1, −1) to (3, 1) and on; (5, −1) to (5, 1) and on; (1, 1) to (3, 2) and on. They all meet at **(5, 3)**. Check: from (5, 3) to (3, 1) is (−2, −2); doubled is (−4, −4), giving (1, −1) ✓.

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9 The canteen manager surveys all 120 students in Year 8 about lunch. Some results are in the two-way table. **[1]**

	Hot meal	Packed lunch	Total
Boys	38		60
Girls		31	
Total	67		120

(a) How many girls have a hot meal?

> **Answer:** 29
> **Notes / working:** Hot meal total is 67 and 38 are boys, so 67 − 38 = **29** girls. (Check: girls total = 120 − 60 = 60, and 29 + 31 = 60 ✓.)

(b) One of the 120 students is chosen at random. Find the probability that the student has a packed lunch. **[1]**

> **Answer:** 53/120
> **Notes / working:** Packed lunch total = 120 − 67 = 53 (or 22 boys + 31 girls). P(packed lunch) = **53/120** (= 0.4416…).

(c) The manager plans some more survey questions. **[1]**
Which question gives continuous data?

- ☐ A) What is your favourite canteen meal?
- ☐ B) How long, in minutes, did you wait in the queue today?
- ☐ C) How many times did you buy a drink this week?
- ☐ D) Which year group are you in?

> **Answer:** B) How long, in minutes, did you wait in the queue today?
> **Notes / working:** Time is measured, and can take any value in a range (e.g. 4.5 minutes), so it is **continuous**. Favourite meal and year group are categorical. ‘How many times’ is counted, so it is discrete.

(d) To find out what the whole school thinks about the canteen, the manager asks the first 30 students in the queue on Monday. **[1]**
Give one reason why this sample may not represent the whole school.

> **Answer:** 1 mark for a sensible reason, e.g. only students who use the canteen are asked (students with packed lunches are left out); students at the front may be from one class or year group; it is only one day; 30 is a small sample; students in the queue probably like the canteen. Do not accept vague answers like 'it is not fair' without saying why.
> **Notes / working:** It is a biased sample: everyone asked is already queuing, so students who bring a packed lunch (and may dislike the canteen) are not asked. Also the first 30 may all be from the same class or year, and it is only one day.

(e) In a survey of 160 students, 52 chose rice as their favourite meal. The results are shown in a pie chart. **[1]**
Work out the angle of the sector for rice.

> **Answer:** 117 °
> **Notes / working:** Each student is 360 ÷ 160 = 2.25°. Rice: 52 × 2.25 = **117°**. (Or 52/160 × 360 = 117.)

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10 Laila makes a spinner with 4 equal sections numbered 1, 2, 3 and 4. She spins it 200 times. It lands on 4 a total of 62 times. **[1]**

(a) Write down the experimental probability of the spinner landing on 4.

> **Answer:** 0.31 or 31/100 or 62/200
> **Notes / working:** Experimental probability = number of times ÷ number of trials = 62/200 = **0.31** (31/100).

(b) If the spinner is fair, how many times would you expect it to land on 4 in 200 spins? **[1]**

> **Answer:** 50
> **Notes / working:** Theoretical probability = 1/4. Expected number = 1/4 × 200 = **50**.

(c) Laila wants a more reliable estimate of the probability of landing on 4. What should she do? **[1]**

> **Answer:** 1 mark for: spin it many more times (e.g. 1000 or more trials), or repeat the experiment and combine the results. Do not accept 'spin it again' without saying more times, or 'make it fair'.
> **Notes / working:** Do **many more spins** (for example 1000). The more trials, the closer the experimental probability usually gets to the true probability. Then compare it with the theoretical 0.25.